

Yale AP PhD Interview

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Coarse-grained quantum state tomography with optimal POVM construction

η_0, η_1 : the probability of **failing** to detect a signal from the $|0\rangle$ and $|1\rangle$ state.

$$\begin{aligned}\Pi_0 &= \eta_0|0\rangle\langle 0| + \eta_1|1\rangle\langle 1| && \text{no detection} \\ \Pi_1 &= (1 - \eta_0)|0\rangle\langle 0| + (1 - \eta_1)|1\rangle\langle 1| && \text{a detection event}\end{aligned}$$

Extending this framework to N -qubit system, the signal detection probability becomes:

$$M_{\text{CG}} = I^{\otimes N} - \Pi_0^{\otimes N}$$

resulting in two positive semidefinite observable operators $\{M_{\text{CG}}, I^{\otimes N} - M_{\text{CG}}\}$.

By implementing unitary gates $G^k(\hat{\theta})$ via parameterized quantum circuits, an extended set of CG POVM operators, $\Omega^k = G^{k\dagger}(\hat{\theta})M_{\text{CG}}G^k(\hat{\theta})$ can be constructed.

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Idea: Uniformly distributed Measurement Operators across Hilbert space.

Uniformness of the set of measurement operators can be represented by von Neumann entropy of Gram matrix

$$S = -\text{Tr}(\Pi' \ln \Pi') = -\sum_i \lambda'_i \ln \lambda'_i$$

where Gram matrix is defined as each element is the inner product of the POVM bases $\Pi_{ij} = \text{Tr} \Omega^i \Omega^j$, and normalized $\Pi' = \Pi / \text{Tr} \Pi$ so that sum of eigenvalues becomes 1.

To maximize von Neumann entropy S , we iteratively update the circuit parameters $\hat{\theta}$. That is, we solved $\arg \max_{\hat{\theta}} S$.

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- (a) The CG POVM based QST process for a two-qubit system is illustrated, where quantum state tomography is performed after an arbitrary quantum operation $\mathcal{O}$ and subsequent projection basis gates $G^k(\theta)$.
- (b) Higher von Neumann entropy of the CG POVM set corresponds to reduced infidelity, demonstrating robustness against the statistical noise.
- (c) CG POVM sets with higher entropy provide more consistent reconstruction results, demonstrating their state-independent performance across a variety of quantum states.

Dynamic Decoupling (DD) sequences, consisting of multi-pulse control on the NV spin, have been developed to achieve entangling gates with selected target nuclear spins while reducing decoherence from the broader spin bath.

An additional selective phase-controlled **radio-frequency (rf)** driving on nuclear spins, referred as DDrf gate, enables elaborated control of individual spins.

$$\begin{aligned}
 \mathcal{H} = & \underbrace{\gamma_c B_z I_z^i + A_{||}^i S_z I_z^i + A_{\perp} S_z I_x^i}_{\text{Drift Hamiltonian}} + \underbrace{\frac{1}{\sqrt{2}} \Omega_{\text{MW}} S_x}_{\text{NV electron spin control}} + \underbrace{2\Omega_{\text{rf}} I_x}_{^{13}\text{C spin control}} \\
 \rightarrow \mathcal{H} = & |0\rangle\langle 0| \otimes \omega_0 I_z + |-1\rangle\langle -1| \otimes (\omega_L I_z - A_{||} I_z - A_{\perp} I_x) \\
 \rightarrow \mathcal{H} = & |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1
 \end{aligned}$$